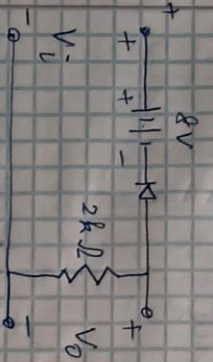
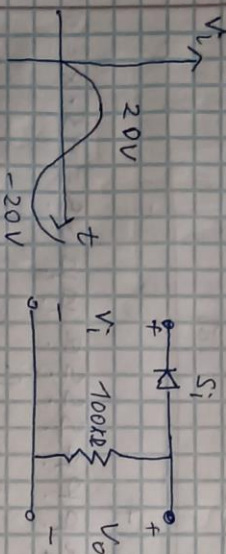


HW # 7

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2) For positive pulse of V_i and $V_o = 0$ Si diode open. For $-20V \leq V_i \leq 0V$ diode "ON" and

$$V_o = V_i + 0.7V$$

For $V_i = -20V$, $V_o = -20V + 0.7V = -19.3V$

For $V_i = 0V$, $V_o = 0.7V + 0.7V = 1.4V$

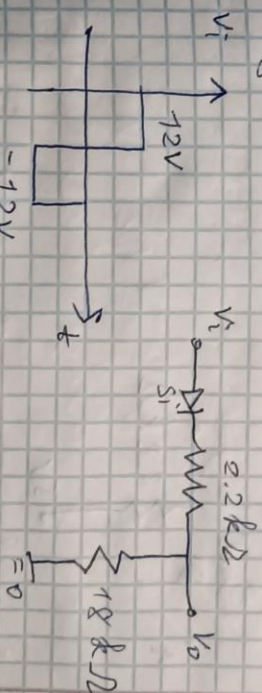
3) For $V_i \leq 8V$ 8V-battery will ensure the diode is forward biased and $V_o = V_i - 8V$

At $V_i = 8V \Rightarrow V_o = 0V$

At $V_i = -20V \Rightarrow V_o = -2.8V$

For $V_i > 8$ the diode is reverse biased and $V_o = 0V$

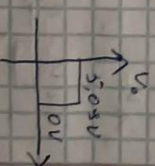
page 12 & 2 x 33



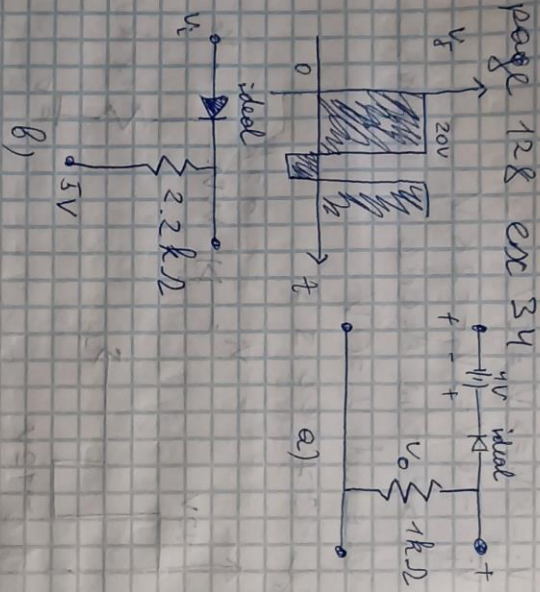
2) Positive pulse of V_i

$$V_o = \frac{1.8k(12 - 0.7)}{1.8k + 2.2k} = 5.08V$$

Negative pulse of V_i diode "OPEN" $V_o = 0V$

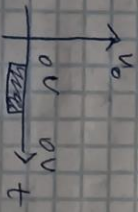


3



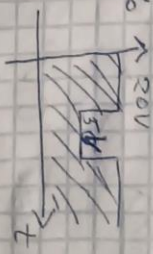
8)

a) $V_i = 20V$ the diode is "reverse biased" and $V_o = 0V$
 For $V_i = -5V$ the diode is "ON"
 KVL: $-5V + 4V - V_o = 0 \Rightarrow V_o = -1V$

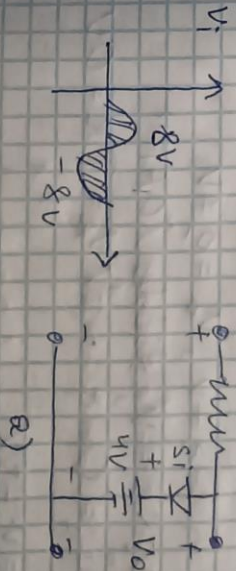


b) For $V_i = 20V$ the diode is "ON"
 short circuit $\Rightarrow V_o = V_i = 20V$

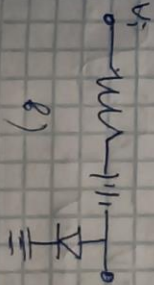
For $V_i = -5V$, both V_i and 5V supply reverse-bias the diode and separate V_i from V_o .
 V_o is connected directly through the $2.2k\Omega$ to the 5V supply and $V_o = 5V$



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8)



5

a) Diode "ON" for $V_1 \geq 4, \neq V$

For $V_1 > 4, \neq V$, $V_0 = 4V + 0, \neq V = 4, \neq V$

For $V_1 < 4, \neq V$, diode "OFF", $V_0 = V_1$



b) diode "ON" for $V_1 \geq 3, \neq V$

but V_0 is voltage across diode

For $V_1 \geq 3, \neq V$, $V_0 = 0, \neq V$

For $V_1 < 3, \neq V$, diode "OFF"

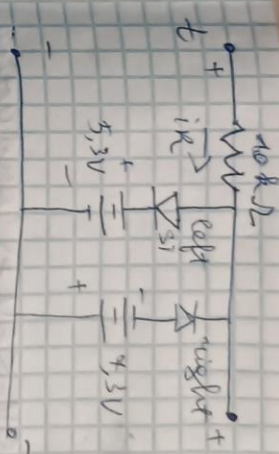
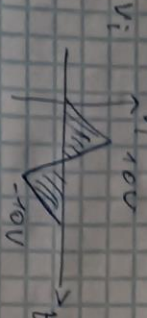
$I_0 = I_R = 0mA$ and $V_R = I_R R = 0V$

$\Rightarrow V_0 = V_1 = 3V$

At $M_1 = 0V$, $V_0 = -3V$

$V_1 = -8V$, $V_0 = -8 - 3 = -11V$

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6

For positive region of V_1 :

right si diode is reverse-biased

left si diode is "ON" for V_1 greater

than $5,3V + 0,7V = 6V$

$V_0 = 6V$ for $V_1 \geq 6V$

For $V_1 < 6V$ both diodes are reverse

baised $\Rightarrow$ For negative region of V_1 :

left si diode is reverse-biased

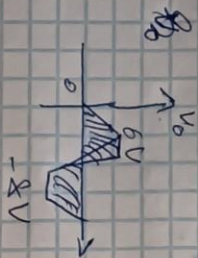
right si diode is "ON" for V_1 more

negative than $7,3V + 0,7V = 8V$

$V_0 = -8V$ for $V_1 \leq -8V$

For $V_1 > -8$ both diodes are re-

verse biased and $V_0 = V_1$



δ_R : For $-8V \leq V_f \leq 6V \Rightarrow$ conduction through the $10\text{ k}\Omega$, so $\delta_R = 0\text{ mA}$

For $V_f \geq 6V \Rightarrow V_R = V_i - V_D = V_f - 6V$

For $V_f = 10V \Rightarrow V_R = 4V$

$$\delta_R = \frac{4V}{10\text{ k}\Omega} = 0.4\text{ mA}$$

For $V_f \leq -8V \Rightarrow V_R = V_i + 8V$

For $V_f = -10V \Rightarrow V_R = -2V$

$$\delta_R = \frac{-2V}{10\text{ k}\Omega} = -0.2\text{ mA}$$

